Vent PROTOTYPE

Manufacturing intelligence, *built for the floor.*

An AI-driven platform that captures operator knowledge in real time and turns it into structured intelligence for QA, manufacturing management, and facility leadership, in a GMP-compliant, regulation-ready system.

Darren Doherty
Founder · Bioprocess Associate, WuXi Biologics

Dogpatch Labs Application
February 2026

THE PROBLEM

The floor knows. The facility doesn't.

In regulated biologics manufacturing, the operators closest to the process are the first to spot deviations, inefficiencies, and procedure gaps. Almost none of it gets captured in time to be useful.

🚨

Reporting is feared

Deviation systems are traceable. Operators know that raising an issue means their name is attached. So observations go unreported, and the facility operates blind.

📋

Documentation consumes the day

30–40% of facility man-hours go to documentation. SOPs live in PDFs nobody reads proactively. Deviations are filed weeks after they happen. The signal is always delayed.

📉

Signal degrades at every handoff

By the time an observation reaches QA or management, it has been filtered, delayed, and summarised. The original insight , the one that could prevent the next deviation , is gone.

> *"The same deviation happens three times across three shifts before anyone connects the dots , because the system isn't designed to connect them."*
>
> , A reality observed daily on biologics manufacturing floors

THE COST

This costs the industry billions.

These aren't theoretical numbers. They are the operating reality of every regulated biologics facility in the world right now.

40–60

man-hours
per CAPA investigation event

€1–5M

per batch failure
in biologics drug substance manufacturing

30–40%

of facility man-hours
consumed by documentation in GMP operations

60–90

days
average deviation-to-closure cycle under traditional QMS workflows

💡 And behind every one of these numbers is a human cost: the operator who noticed something and said nothing, the QA reviewer buried in paperwork, the director making decisions on data that's already two weeks old.

THE SOLUTION

Vent is an intelligence amplifier.

It takes the deep, real-time knowledge that exists in the heads and hands of the people closest to the process , and makes that knowledge instantly actionable at every level of the organisation.

WITHOUT VENT

- ✗ Operators swallow observations to avoid blame
- ✗ SOPs sit in binders nobody reads in the moment they're needed
- ✗ QA manually cross-references deviations against SOP libraries
- ✗ CAPA packages built from scratch, 40+ hours each
- ✗ Recurring patterns invisible until a third or fourth occurrence
- ✗ Director gets a weekly summary written by someone else

WITH VENT

- ✓ Operators submit anonymously in 30 seconds, from any device
- ✓ SOP query in plain language , step-by-step answer in seconds
- ✓ AI cross-references every submission against all relevant SOPs automatically
- ✓ Structured CAPA package drafted in seconds, reviewed by QA
- ✓ Pattern detection flags recurring issues on first repeat occurrence
- ✓ Director sees live facility intelligence , not a weekly summary

HOW IT WORKS

Four layers. One closed loop.

1

Capture

OPERATOR

Anonymous 30-second observation submission. Plain language. Photo/video attachments. No form templates, no deviation numbers, no fear of blame.

→ Raise an observation
→ Query a procedure
→ Build a document

2

Analyse

AI ENGINE

Claude + VoyageAI RAG pipeline. Semantic SOP search, root cause hypothesis, priority classification, CAPA generation, contact routing, pattern detection.

→ SOP cross-reference
→ Risk level assessment
→ Pattern matching

3

Workflow

QA & MFG SUPPORT

Structured 4-phase lifecycle. Electronic signatures on approval transitions. Immutable audit trail. SOP change management. 21 CFR Part 11 / EU Annex 11 compliant.

→ QA review & sign-off
→ CAPA tracking
→ Doc control

4

Intelligence

LEADERSHIP

Live facility dashboard. Submission trends by area, priority, and status. CAPA metrics. Cross-submission pattern analysis. Real decisions, not weekly summaries.

→ Director dashboard
→ Trend analysis
→ Compliance reporting

Every submission tightens the loop. Every observation makes the facility smarter.

THE DATA MOAT

The longer Vent runs, the harder it is to replace.

Every submission enriches the facility's pattern library. After six months of live data, Vent knows that site's deviation patterns better than any QA system ever has. You can copy software. You cannot copy data.

The compounding advantage

Each observation submission enriches the facility's pattern library. Cross-submission detection sharpens with every data point. Six months in, Vent understands that site's deviation fingerprint better than any human analyst could.

Why competitors cannot catch up

A competitor arriving at a facility where Vent has been running for a year has no access to 12 months of anonymised observations, SOP gap flags, and deviation patterns. That intelligence is locked in Vent's model of that site, and it took a year of real operations to build.

> Every site that runs Vent for a full production cycle becomes a facility intelligence asset, one that compounds every shift and makes displacement progressively harder to justify.

PRODUCT , OPERATOR LAYER

Designed for the floor, not the office.

Two core tools, built for people whose hands are busy. No training required. No system navigation. Just ask or report.

vent-production.up.railway.app/query

Query SOPs | Doc Builder | Operator

Ask the *procedures.*

Search the facility SOPs in plain language. Get step-by-step instructions and exact parameter values.

pH control | Inoculation criteria | DO control cascade | Bleed rate

Ask a question about any SOP procedure, parameter, or step...

, Raise an observation

Raise an observation

Anonymous · routed to QA & MSAT

OBSERVATION

What did you observe? Any deviation, concern, or SOP gap?

Upstream Bioreactor | Low | Med | High

Anonymous to colleagues. Traceable by the system for compliance.

PHOTOS OR VIDEO , OPTIONAL

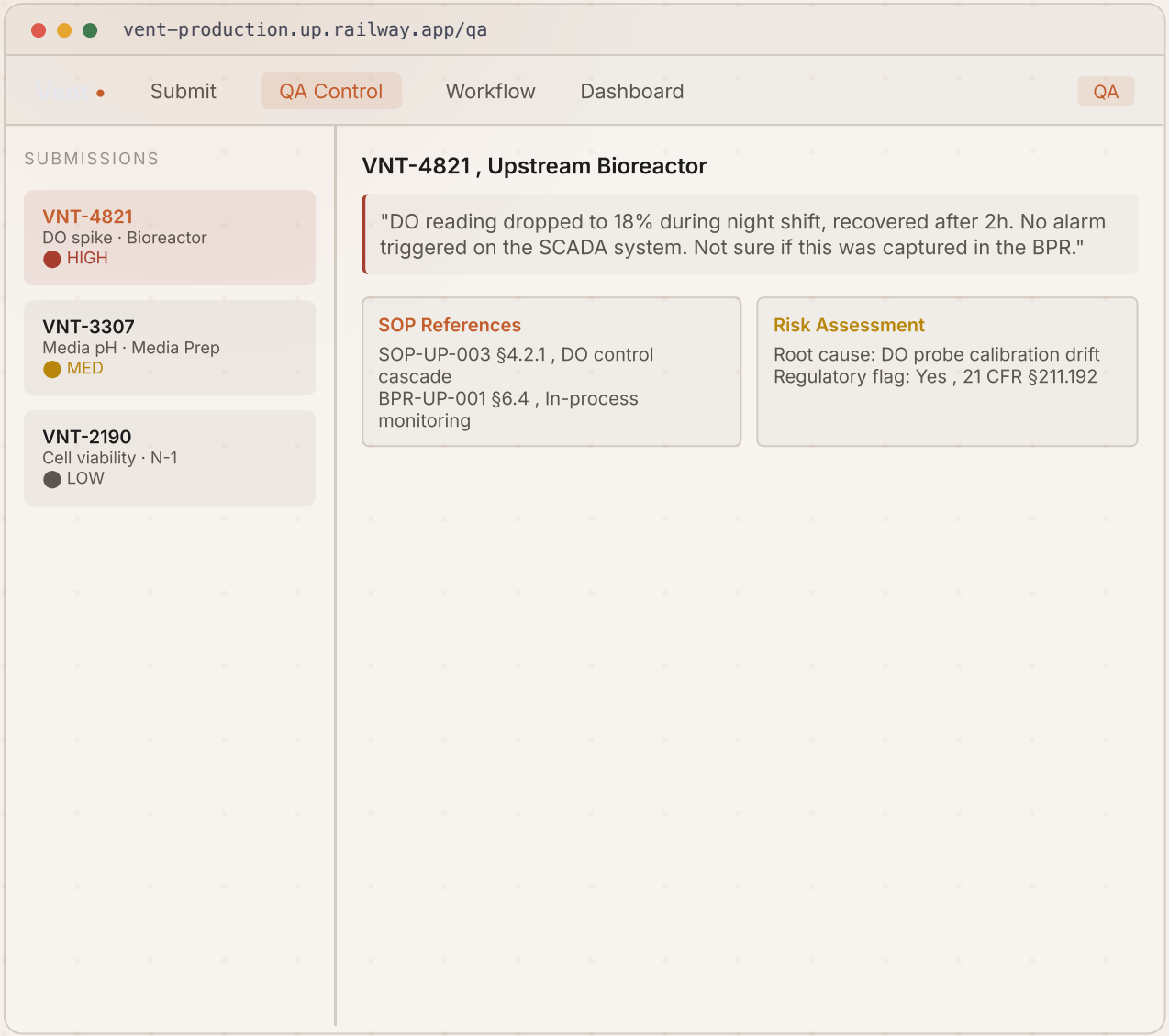
Add photos or video
Drag & drop or click · up to 5 files

Submit observation →

PRODUCT , QA INTELLIGENCE LAYER

QA has a command centre, not a queue.

Every submission arrives fully analysed , SOP references, risk level, corrective actions, and contact routing already generated. QA reviews, decides, signs.



Auto SOP cross-reference

Every submission arrives with the exact SOP sections cited by ref number. Click any reference to read the full section inline.

CAPA package drafted

Corrective actions generated with timing bands , immediate, short-term, long-term , and automatically assigned to the right contact.

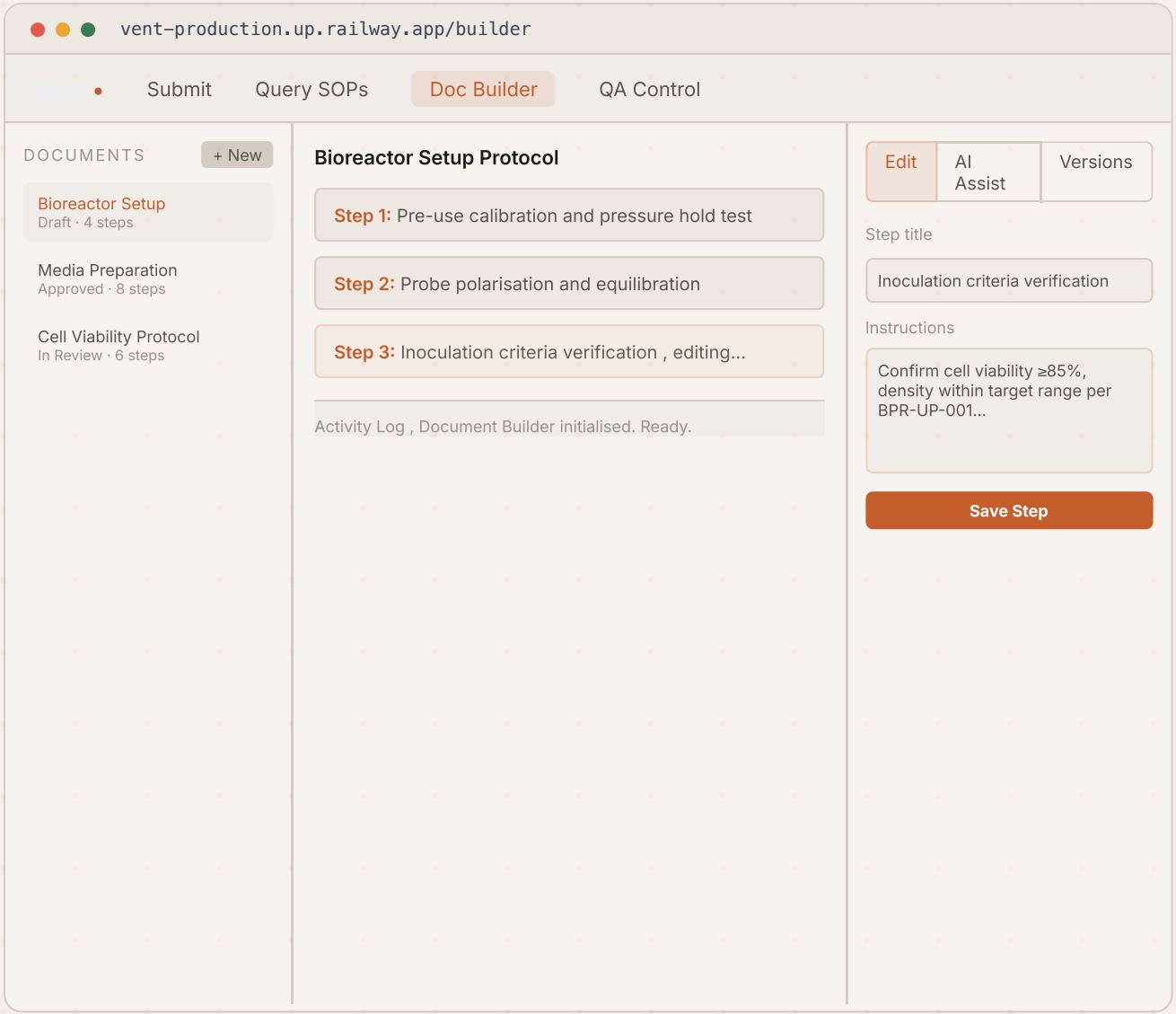
Electronic signatures

QA approval, director sign-off, and closure require user ID + reason , logged with timestamps and SHA-256 checksums.

PRODUCT , DOCUMENT INTELLIGENCE LAYER

Living documents, authored by the people who use them.

A full IDE-style authoring environment for SOPs, work instructions, and batch record templates , built into the same platform operators use every day.



Key capabilities

- ✓ Step-based document structure with version snapshots
- ✓ Drag-to-reorder steps , VS Code-style IDE layout
- ✓ Draft → In Review → Approved lifecycle tracking
- ✓ AI Assist tab , Claude-powered draft generation
- ✓ Real-time activity log for audit compliance
- ✓ Media attachments , photos, video, files

Operators author documents in the same tool they use for queries and observations. No EDMS login, no separate system, no friction.

COMPLIANCE ARCHITECTURE

Regulation-ready by design.

Most QMS tools bolt compliance on as an afterthought. Vent is built for regulated environments from the first line of code , because it was designed by someone who lives those requirements.

21 CFR PART 11 / EU ANNEX 11

Electronic signatures with user ID and justification required on all approval transitions , QA sign-off, director approval, closure, rejection.

ROLE-BASED ACCESS CONTROL

Six distinct roles (Operator, QA, Director, MSAT, Engineering, Admin) with fine-grained route-level enforcement. Sensitive transitions locked by role.

IMMUTABLE AUDIT TRAIL

Every action , submission, status change, SOP modification, CAPA update, login , logged with SHA-256 checksums, timestamps, IP addresses, and user agents.

DATABASE-LEVEL SECURITY

Supabase Row Level Security policies. No direct table access without authentication. All queries proxy through the authenticated Express server.

ALCOA+ DATA INTEGRITY

Data is Attributable, Legible, Contemporaneous, Original, and Accurate. Anonymity for operators does not compromise traceability for the system , these are separated by design.

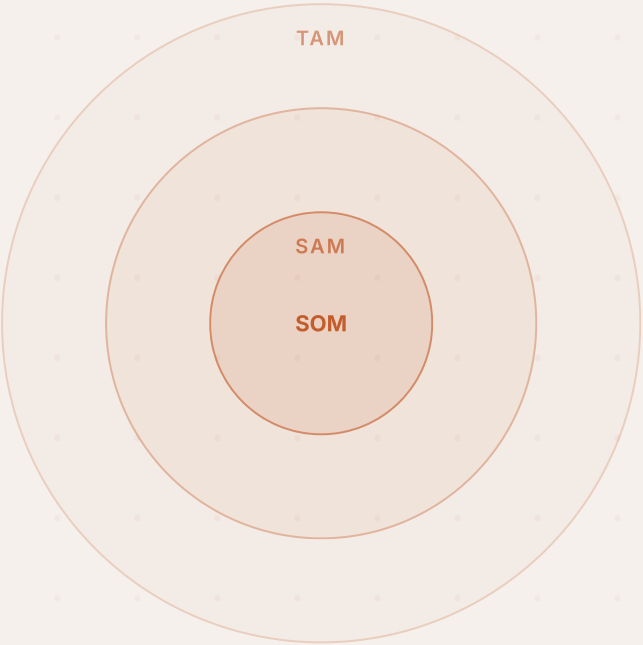
Why this matters for customers

FDA and HPRA inspectors audit electronic records systems. A platform without a compliant audit architecture cannot enter a GMP facility , full stop. Vent is inspection-ready from day one, which removes the single biggest procurement blocker.

MARKET OPPORTUNITY

A regulated, underserved, fast-growing market.

Biologics is the fastest-growing segment of pharma. Every facility runs on GMP. Every GMP facility has the same floor-to-QA information problem. None of them have a solution built for the floor.



TAM , Global Quality Management Software — **$11B**

2024 market size · 10.6% CAGR · life sciences, biotech, medtech and regulated manufacturing

SAM , Pharma QMS, CAPA & Doc Control — **$3B**

Pharma QMS 2030 projection · 13.3% CAGR · ~5,000 regulated biologics target sites globally

SOM , AI-native Floor Intelligence (5yr) — **$200M**

~4,000 GMP sites (EU + US) · avg €50K ARR · biologics and drug substance manufacturing

Ireland as the entry point

9 of the world's top 10 pharma companies manufacture in Ireland. Pfizer, Eli Lilly, MSD, BMS, Regeneron, AbbVie — the world's densest pharma cluster per capita. The founder works inside it every day.

WHY IRELAND FIRST

Europe's most concentrated biopharma cluster.

9 of the world's 10 largest pharmaceutical companies manufacture in Ireland. This is not a market we need to reach — it is the environment the founder operates in every day.

CORK / MUNSTER CLUSTER

- **Pfizer** Ringaskiddy, Cork
- **Eli Lilly** Kinsale, Cork
- **MSD** Brinny, Cork
- **GSK** Dungarvan, Waterford
- **Regeneron** Limerick

NATIONAL SITES

- **AbbVie** Sligo
- **Alexion / AZ** Athlone
- **Takeda** Bray, Wicklow
- **BMS** Dublin 15
- **WuXi Biologics** Dundalk, Louth

9/10

of the world's top 10 pharma companies manufacture here

€80B+

pharma exports annually — Ireland's #1 export sector

90+

licensed manufacturing sites across the island

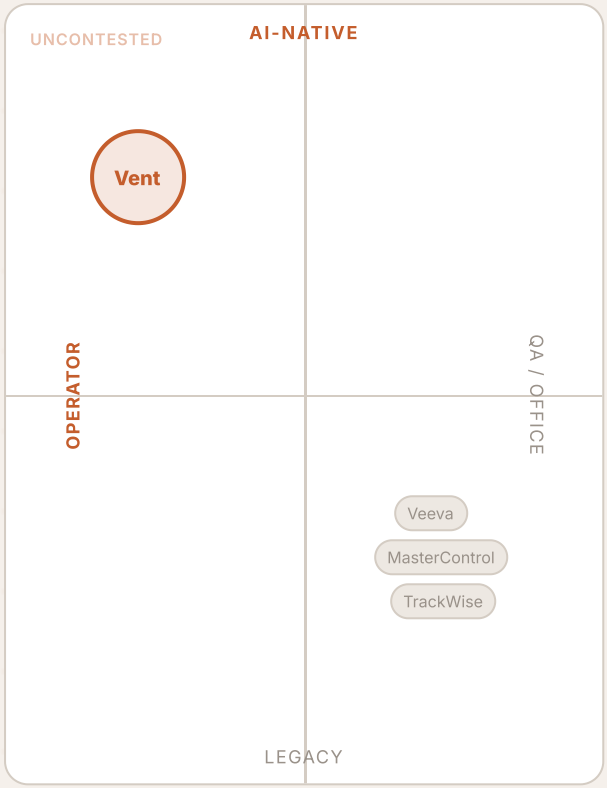
INSIDER ACCESS

The founder works inside one of these facilities. The first sale is a warm door, not a cold call.

COMPETITIVE LANDSCAPE

The incumbents were built for the office. Vent is built for the floor.

Existing QMS tools serve QA teams who control the budget. Nobody has ever built for the operators who generate the signal. That gap is where Vent lives.



	Veeva	MasterControl	TrackWise	Vent
Anonymous floor reporting	✗	✗	✗	✓
AI SOP analysis (RAG pipeline)	✗	✗	✗	✓
Auto-generated CAPA packages	✗	✗	✗	✓
Operator-first floor UX	✗	✗	✗	✓
Deviation & CAPA management	✓	✓	✓	✓
Document control	✓	✓	✗	✓
Training LMS (same platform)	add-on	✓	✗	✓
Flat site-based pricing	✗	✗	✗	✓
Typical annual cost per site	**€60–240K**	**€50–300K**	**€60–480K**	**€35–60K**

TO REPLICATE VENT TODAY

QMS licence + doc management + training LMS + AI integration layer

Combined cost **€100–500K+/yr**

Vent (all-in) **€35–60K/yr**

BUSINESS MODEL

Site-based SaaS. Simple to land, strong to expand.

Annual per-site licences, with natural expansion across departments and up the management stack. CDMOs operating multiple sites are the highest-value enterprise accounts.

OPERATOR

€15K

per site / per year

- ✓ SOP Query (RAG-powered)
- ✓ Anonymous observation submission
- ✓ Document Builder
- ✓ Notification system

Entry point for floor teams

QA + WORKFLOW

€35K

per site / per year

- ✓ Everything in Operator tier
- ✓ QA Control Centre
- ✓ CAPA tracking & management
- ✓ SOP change management + audit trail
- ✓ Electronic signatures (21 CFR Part 11)
- ✓ 4-phase workflow routing

Core product , where value is captured

ENTERPRISE

€60K+

per site / per year

- ✓ Everything in QA tier
- ✓ Multi-site analytics & benchmarking
- ✓ Director intelligence dashboard
- ✓ API access + EDMS integration
- ✓ Custom SOP library ingestion
- ✓ Dedicated onboarding & validation support

CDMOs and multi-site operations

WHERE WE ARE

Pre-revenue. Post-prototype. Honest picture.

The product exists, it works end-to-end, and it was built by someone who lives the problem every day. No co-founder, no investment, no customers yet. Here is what is true.

Live

Working product deployed on Railway. Full stack: submission → AI analysis → CAPA → QA workflow → director dashboard. You can demo it today.

Solo

Designed, built, and deployed by a single founder, a practising bioprocess associate, in evenings and weekends while working full time on the manufacturing floor. Zero external funding.

Insider

The problem is not theoretical. The founder observes the floor-to-QA information gap every working day. Built from daily lived experience, not user interviews.

Part 11

21 CFR Part 11 compliant audit trail, e-signatures, and SHA-256 checksums built from day one. The compliance architecture passes inspection without further development.

What exists today

- ✓ End-to-end submission → AI analysis → CAPA workflow
- ✓ QA Control Centre with SOP cross-reference
- ✓ Document Builder with version control and AI Assist
- ✓ 21 CFR Part 11 compliant audit trail and e-signatures

What's next (with Dogpatch)

- → First paying pilot (CDMO or MNC, Ireland)
- → IQ/OQ/PQ validation package for GMP deployment
- → Visual SOP Engine (Phase 2)
- → Technical co-founder / first hire

THE FOUNDER

Domain expertise is the moat.

DD

Darren Doherty

Founder, Vent

Bioprocess Associate
WuXi Biologics

BSc Immunology (Scholar)
Trinity College Dublin

> *"I am not a software person who discovered pharma. I am a pharma person who built software to solve a problem I live every day."*

FLOOR EXPERTISE

Works directly with operators on fed-batch and perfusion upstream processes. Knows their problems intimately , not from research, from daily experience.

CROSS-FUNCTIONAL

Works across QA, Manufacturing Support, and Engineering in cross-functional teams. Understands every stakeholder's position from the inside.

REGULATORY LITERACY

Understands the IQ/OQ/PQ validation requirements and audit trail standards needed to deploy software in a GMP environment. Vent is built to pass inspection.

LOOKING FOR

Technical co-founder (software/AI background). Advisory board with pharma commercial and regulatory expertise. Senior operators as design partners.

THE ASK

What we're asking for.

The product exists. The problem is real. The market is here. What I need is the platform to build this properly , and the right people in the room.

🏢

A place to build

A desk in Dogpatch so I can work on Vent full time , without splitting focus between the manufacturing floor and the laptop. This is the most important thing.

💵

Early-stage funding

A small seed grant (€25–50K) to cover infrastructure costs (AI APIs, hosting, database), free myself from employment, and reach a first paying pilot , without burning runway on anything else.

🤝

Network & mentorship

Access to Dogpatch's pharma and medtech connections in Ireland. Ireland is home to the world's densest biologics manufacturing cluster. An introduction to the right QA director or site head is worth months of cold outreach.

What you get

Early access to a founder who works inside the industry, knows every person on every floor, and has already built and deployed the product that solves a problem every GMP facility pays to manage badly. The moat is knowledge, and it compounds every day I stay on the floor.

- ✓ Live product you can demo today
- ✓ Daily floor access to the exact problem being solved
- ✓ Domain expertise no generalist founder can replicate
- ✓ A market that will only grow as AI enters regulated industries